

Multi-View Comparative Self-Supervised Graph Learning for Neuro-disease Diagnosis Using Functional Brain Networks

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Abstract—The burgeoning interest in diagnosing neurological diseases through the analysis of functional brain networks (FBNs) has been a focal point of recent research. Despite the promise of this approach, conventional graph learning techniques are often constrained by their reliance on extensive datasets annotated with clinical information. To transcend this constraint, we introduce a novel self-supervised graph learning model, termed Multi-View Comparative Self-Supervised Graph Learning (MCS-GL), which leverages Personalized PageRank (PPR) to increase samples and uses local-global information as multi view for comparative learning to enhance FBN-based diagnostics. With this approach, a pre-trained model is designed to generate information-rich Functional Magnetic Resonance Imaging (fMRI) representations from unlabeled training data. Fine-tuning the model for downstream disease identification tasks significantly reduces the requirement for large amounts of labeled training data. We have rigorously evaluated the efficacy of our MCS-GL model through its application to an autism diagnosis task. The experimental outcomes not only substantiate the superiority of our model but also underscore the pivotal role of self-supervised learning in advancing FBN-based diagnostics of neurological diseases.

Index Terms—Autism Disorders, Functional Brain Networks, Self-supervised Learning, Graph Convolutional Neural Networks, Neuro-Disease Diagnosis

I. INTRODUCTION

In recent years, the field of neurodiagnostics has witnessed a surge of interest in the cerebral network-based approaches

for the identification and characterization of neurological disorders [1-4]. Functional brain network analysis is now recognized as a crucial tool for studying the interplay between brain structure and function. This sophisticated analysis facilitates the identification of functional connectivity patterns across various brain regions; it also uncovers the intricate links between these patterns and cognitive, behavioral, as well as neurological diseases [5, 6]. Examples include identifying the Alzheimer's disease [7, 8], Autism [9], and Parkinson's disease [10].

While traditional methods, such as Support Vector Machines (SVMs) [11], Random Forests [12], and Graph Convolutional Neural Networks (GCNNs) [13], have achieved notable success in leveraging machine learning for neuro-disease diagnosis, they often face limitations due to the requirement for substantial volumes of labeled data. The high demand for acquiring neuroimaging data, coupled with small sample sizes and individual differences, makes it difficult for these methods to reach their full potential. Therefore, there is a strong desire to reduce the need for data volume while enhancing the generalization ability of the model.

In contrast, the potential of graph self-supervised learning methods has been amply demonstrated across various domains. For instance, Deep Graph Infomax (DGI) is successful in maximizing the mutual information between local and global representations [14]. Similarly, Graph Contrastive Learning (GraphCL) [15] has significantly enhanced the quality of graph representations by leveraging data augmentation and by maximizing consistency across different graph views. Graph Contrastive Coding (GCC) [16] has also excelled in capturing salient information within the graph structure, achieving commendable performance, particularly notable for its reduced reliance on extensive labeled data

Drawing inspiration from these advances, we have architected a novel brain network-oriented graph supervision model. Our proposed framework, as depicted in Fig. 1, is

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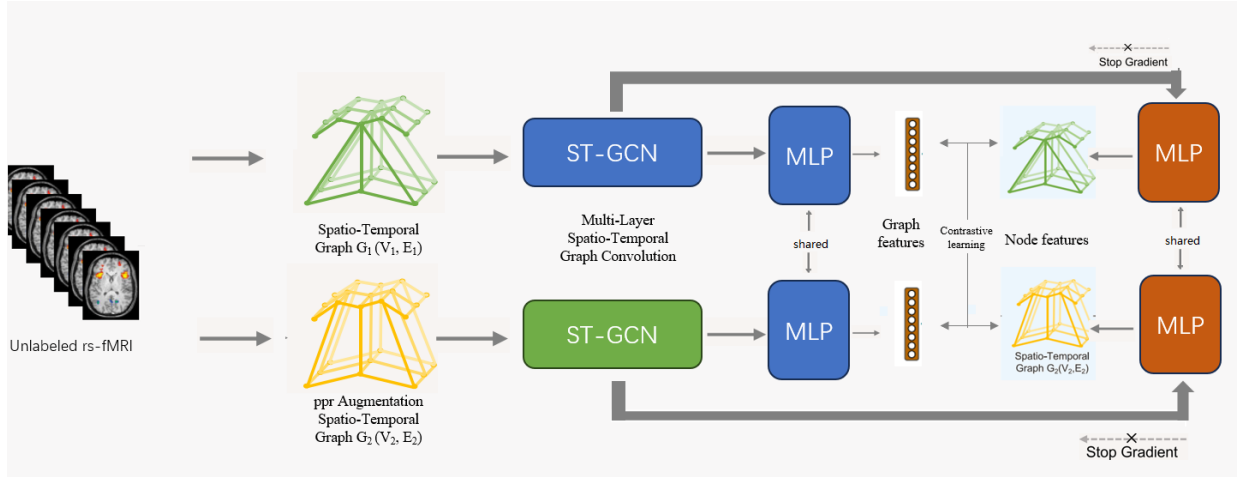


Fig. 1. The pre-training process of the proposed MCS-GL model is illustrated, which consists of two structures: 1) increasing the sample size with ppr augmentation. 2) Two fMRI encoders that do not share parameters are used to learn the representation from the augmented data in an unsupervised manner.

designed for the detection of fMRI-based degenerative neurological diseases. It comprises two principal components: (1) a network pre-task that leverages local and global information for unsupervised training on unlabeled data; and (2) a network downstream task, where we transfer models from the pre-task for fine-tuning and training on labeled data.

Specifically, our MCS-GL employs a 3-layer spatio-temporal graph convolutional layer (ST-GCN) which can extract spatial and temporal information to capture the dynamic interaction patterns between brain regions to distill both global and local features for comparative learning. Our model's performance has been rigorously validated through experiments on major depressive disorder (MDD), with results indicating significant improvements in classifying degenerative neurological diseases. By integrating data enhancement techniques and crafting a bespoke loss function for contrastive learning, we have excelled in downstream tasks.

The MCS-GL model's pre-training regimen is meticulously detailed, highlighting its dual pivotal architectural elements: (1) expansion the sample size using PPR sampling; and (2) two distinct fMRI encoders that operate in an unsupervised manner without shared parameters, learning representations from augmented data. Each feature extraction module integrates a spatial-temporal graph convolutional network (ST-GCN) for feature learning, and employs two types of multilayer perceptrons (MLPs) designed to align global features with local ones. A comparative learning loss function is leveraged to optimize the congruence of representations extracted from identical subjects. To encapsulate, the innovation of this paper are summarized as follows:

1. We present the pioneering application of PPR sampling enhancement to functional brain network(FBN) data, marking a significant advancement in the field of neuroimaging analysis.
2. We introduce self-supervised learning methodologies into the diagnostic paradigm of neurological disorders, effectively mitigating the

challenges associated with limited sample sizes and enhancing the robustness of our diagnostic models.

II. DATA PROCESSING AND METHODOLOGY

A. Data processing

In the REST-MDD study, focusing on Resting-state Magnetic Resonance Imaging (REST-MDD), we opted to validate our model using data from the three most extensive sites: site 20, site 21, and site 1. The patient demographics at these sites were as follows: site 20 had 282 individuals with Major Depressive Disorder (MDD) and 251 healthy controls (HCs); site 21 had 86 MDD patients and 70 HCs; and site 1 had an equal number of MDD patients and HCs, with 74 in each group. For site 20, resting-state functional magnetic resonance imaging (rs-fMRI) data were collected using a Siemens Tim Trio 3T scanner with a 12-channel receiver coil. For each fMRI, the repetition time (TR) was 2000ms, the echo time (TE) was 30ms, the voxel size was $3.44 \times 3.44 \times 4.00 \text{ mm}^3$, the number of slices was 32, the slice thickness was 3.0mm, the gap was 1.0mm, the number of time points was 242, the field of view (FOV) was 220×220 and the flip angle was 90 degrees. For site 21, data were acquired with a Siemens Tim Trio 3T scanner with a 32-channel receive coil. For each fMRI, the TR was 2000ms, the TE was 30ms, the voxel size was $3.12 \times 3.12 \times 4.20 \text{ mm}^3$, the number of slices was 33, the slice thickness was 3.5mm, the gap was 0.7mm, the number of time points was 240, the FOV was 200×200 , and the flip angle was 90 degrees. For site 1, data were collected using a Siemens Tim Trio 3T MRI scanner with a 32-channel receive coil. For each fMRI, the TR was 2000ms, the TE was 30ms, the voxel size was $3.28 \times 3.28 \times 4.80 \text{ mm}^3$, the number of slices was 30, the slice thickness was 4.0mm, the gap was 0.8mm, the number of time points was 210, the FOV was 212×210 , and the flip angle was 90 degrees. The fMRI data for REST-MDD underwent a rigorous standardization process using the Data Processing Assistant for Resting State Functional Magnetic Resonance Imaging (DPARSF). Initially, the first 10 volumes were discarded to stabilize the magnetization state. The subsequent steps included slice time correction, head motion correction,

and bandpass filtering within the 0.01-0.1Hz frequency range to refine the data quality. We further refined the data by conducting regression analysis to eliminate interference from head motion parameters, white matter signals, and cerebrospinal fluid signals. For precise alignment of T1-weighted images with the functional images, a linear transformation with six degrees of freedom was applied, followed by spatial normalization to convert the data from individual native space to the standardized Montreal Neurological Institute (MNI) space. To conclude the preprocessing, we employed the automated anatomical labeling (AAL) atlas technique to segment the brain into 116 distinct regions of interest (ROIs), from which we extracted the average time series for further analysis.

B. fMRI Augmentation

Causing the difficulty in obtaining fMRI data and preventing overfitting problems in model training we used to augment the data by generating a diffusion matrix of the graph to make the model more expressive as well as have better generalization by comparing the original view with the augmented view.

$$S = \sum_{k=0}^{\infty} \theta_k T^k \in \mathbb{R}^{N \times N} \quad (1)$$

Diffusion is formulated as Eq. (1) where $T \in \mathbb{R}^{n \times n}$ is the generalized transition matrix, θ is the coefficient weights of local-global information [17], Imposing $\sum_{k=0}^{\infty} \theta_k = 1$, $\theta_k \in [0, 1]$.

Given an adjacency matrix $A \in \mathbb{R}^{N \times N}$ and a diagonal degree matrix $D \in \mathbb{R}^{N \times N}$, PPR, i.e., a instantiations of the generalized graph diffusion, are defined by setting $T = AD^{-1}$, and $\theta_k = \alpha(l - \alpha)^k$, where α denotes teleport probability in a random walk [18]. PPR diffusion are formulated in Eq. (2).

$$S^{PPR} = \alpha \left(I_n - (1 - \alpha) D^{-\frac{1}{2}} A D^{-\frac{1}{2}} \right)^{-1} \quad (2)$$

C. Encoder

We define $\mathcal{G} = \{X, A\}$, $X = \{x_1, \dots, x_N\}^T \in \mathbb{R}^{N \times T}$ as the fMRI time series, where $x_i = \{x_{i,1}, \dots, x_{i,T}\} \in \mathbb{R}^T$ is the average value of BOLD in the i_{th} ROIs Region of Interest, T is the timestamp of the BOLD signal, and the number of ROIs is $N=116$, $A \in \mathbb{R}^{N \times N}$ is the adjacency matrix of the graph, all the samples share a common adjacency matrix, and if there is a connection between a point i and a point j , then $a_{ij}=a_{ji}=1$; Otherwise $a_{ij}=a_{ji}=0$.

In traditional graph convolutional networks [19], for a given input $\mathcal{G} = \{X, A\}$ for the features that have been l_{th} layer convolution, in the calculation of the $l + 1$ layer of graph convolution we have

$$X_{l+1} = \sigma(\tilde{A} X_l W_l) \quad (3)$$

where $\sigma(\cdot)$ is the activation function, W_l is the weight of the l -layer graph convolution, and $\tilde{A}$ is the Laplace matrix.

$$\tilde{A} = \hat{A} \hat{D}^{-1} \in \mathbb{R}^{N \times N} \quad (4)$$

where $\hat{D}$ is the degree matrix of $\hat{A} = A + I_N$, I_N is the unit matrix.

However, we only considered the spatial topology in the functional brain network in the process of graph convolution, while ignoring the features in the temporal dimension of fMRI. Therefore we decided to use spatio-temporal convolution to encode the raw data.

$$B(x_{n,t_i}) = \left\{ x_{m,q} \mid d(x_{n,t_i}, x_{m,q}) \leq K, \left| q - t_i \leq \left\lfloor \frac{L}{2} \right\rfloor \right\} \quad (5)$$

We use Eq. $B(x_{n,t_i}) = \left\{ x_{m,q} \mid d(x_{n,t_i}, x_{m,q}) \leq K, \left| q - t_i \leq \left\lfloor \frac{L}{2} \right\rfloor \right\}$ (5) to denote the spatio-temporal neighborhood of x_{n,t_i} , where K denotes the spatial convolution kernel size, L denotes the temporal convolution kernel, and $d(\cdot, \cdot)$ denotes the distance between the two nodes, then we use

$$f_{st}(x_{n,t_i}) = \frac{1}{Z_{n,t_i}} \sum_{x_{n,t_i} \in B(x_{n,t_i})} x_{n,t_i} \cdot W_{n,t_i} \quad (6)$$

to denote the operation of performing spatio-temporal convolution, which Z_{n,t_i} is a regularization factor and W_{n,t_i} is a spatio-temporal convolution kernel. We assume that the input features are $x_{n,t_i} \in \mathbb{R}^{N \times T}$, denoting the t time steps of the N ROIs, and have an adjacency matrix A about them, and then we approximate the spatio-temporal convolution kernel $W(\cdot)$ by decomposing it into a spatial convolution kernel $W_s \in \mathbb{R}^{T \times M}$ and a temporal convolution kernel $W_T \in \mathbb{R}^{M \times \tilde{T}}$; here for the spatial convolution operation we use the traditional graph convolution for features that have not been augmented by the data as in Eq. (3), and use Eq. (7) for features that have been augmented by the data.

$$X_{l+1} = \sigma(S^{PPR} X_l W_l) \quad (7)$$

After spatial convolution, a standard one-dimensional convolution kernel was used to perform the convolution operation on the temporal features on each ROIs: $x_{n,t_i} \otimes W_T \in \mathbb{R}^{N \times \tilde{T}}$ where $\otimes$ denotes the convolution operation.

In our experiments, we use three layers of time-space convolution to encode the input data; we pass the original data (x_{n,t_i}, A) and the augmented data $(x_{n,t_i}, \tilde{A})$ through two encoders, respectively, and $\phi(\cdot)$, $\psi(\cdot)$ map the input from the original feature space $(\mathbb{R}^{N \times T}, \mathbb{R}^{N \times N})$ to the deep feature space $(\mathbb{R}^{N \times \tilde{T}}, \mathbb{R}^{N \times N})$, to obtain (x'_{n,t_i}, A) and $(\tilde{x}'_{n,t_i}, \tilde{A})$, where the two encoders $\phi(\cdot)$, $\psi(\cdot)$ do not share parameters. After that, we add an average pooling layer $\Omega(\cdot): \mathbb{R}^{N \times \tilde{T}} \rightarrow \mathbb{R}^T$ to extract the global information, which aggregates all the learned features on the nodes to obtain a representation of the graph $\vec{x}_g, \vec{\tilde{x}}_g$, followed by two linear projection layers for local and global information $\phi_1(\cdot): \mathbb{R}^{N \times \tilde{T}} \rightarrow \mathbb{R}^{N \times h}$ and $\phi_2(\cdot): \mathbb{R}^T \rightarrow \mathbb{R}^h$ which combine the representation of the points obtained by the encoding layer $\tilde{x}'_{n,t_i}, \tilde{x}'_{n,t_i}$ and the global average pooling representation of the graph $\vec{x}_g, \vec{\tilde{x}}_g$, and the projection layer to make the local and

global features have the same dimensions, obtaining the local and global features with the same dimensions. The local features $\vec{Z}_d^\alpha, \vec{Z}_d^\beta$ and global features $\vec{Z}_g^\alpha, \vec{Z}_g^\beta$ are obtained by projecting the local and global features with the same dimension, α denoting the original data distribution and β denoting the enhanced data distribution. After that, we support the training of downstream tasks by summing up the global features $\vec{Z}_g = \vec{Z}_g^\alpha + \vec{Z}_g^\beta$.

D. Training Model

In order to train the model from end-to-end and give downstream tasks to learn rich node-to-graph representations, we use cosine similarity to measure the distance between nodes and graphs. We define the objective function as follows:

$$\max_{\phi, \psi, \phi_1, \phi_2} \frac{1}{|g|} \sum_{g \in Q} \left[\frac{1}{|g|} \sum_{i=1}^{|g|} \cos(\vec{Z}_d^\alpha, \vec{Z}_g^\beta) + \cos(\vec{Z}_g^\alpha, \vec{Z}_d^\beta) \right] \quad (8)$$

E. Model Fine-tuning

The primary objective of this paper is to fine-tune the model by analyzing target data and transferring the encoder trained on unlabeled, large-scale fMRI data from the source domain to a downstream classification task. Our fine-tuned model comprises several key components: an ST-GCN encoder, a projection layer that converts feature data into graph data, a multilayer perceptron (MLP) with two connected layers, and a Softmax layer for predictions. During this stage, we fine-tune the model to the fMRI data in the target domain. Instead of incorporating the BOLD signal, we directly create a spatio-temporal map for each subject. We then utilize the ST-GCN for feature extraction, which is followed by classification through the projection layer and the MLP. The parameters of the coding layer are obtained from the pre-training step, along with the parameters of the projection layer and the Softmax function. Finally, we update the parameters using the cross-entropy loss function in a supervised learning manner.

F. Complementary Details

Within the proposed MCS-GL framework, the length of the original BOLD signals in the temporal dimension (with original T time points) is 200 and the threshold of connectivity between brain regions is 0.2 and the diffusion index of the diffusion matrix is 0.3 each ST-GCN feature encoder comprises 3 ST-GCN units. For each ST-GCN, the unit commences with a single input channel, denoted as C , while each subsequent layer outputs through 64 channels, represented by M . Spatial convolution is executed with a kernel size K of 1, encompassing a total of 64 kernels. Temporal convolution employs a kernel size L of 11, operates with a stride of 1, and incorporates a dropout rate of 0.5 to prevent overfitting.

The pretext model's optimization leverages the Stochastic Gradient Descent (SGD) algorithm, a standard in contrastive learning approaches. The optimization journey begins with a learning rate set at 0.7, a weight decay of 0.0001, and a

momentum factor of 0.9. Training is conducted in mini-batches of 16 samples, iterating through a total of 50 epochs.

In transitioning to the fine-tuning phase for the Unsupervised Cross-Domain Graph Learning (UCGL) model, the network architecture and initial parameters are preserved from the pretext model's ST-GCN component. Each Multi-Layer Perceptron (MLP) predictor is structured with three fully-connected layers, with neuron counts of 32, 64, and 64, respectively. Post the first fully connected layer, Batch Normalization (BN) and ReLU activation functions are applied to enhance learning dynamics, while the third fully connected layer forgoes BN and ReLU to allow for direct prediction outputs via a Softmax layer.

For the fine-tuning stage, we adopt the Adam optimizer in conjunction with a cosine annealing schedule to refine the learning process. The learning rate is initialized at 0.07, with mini-batches consisting of 16 samples each. The training is set to span 30 epochs to achieve convergence.

III. EXPERIMENT

In this section, we present some details of the experimental setup with several common methods, give the results and analyze them.

TABLE I. Tested cross-site categorization: model from site 20 to site 21 on REST-MDD.

method	ACC	AUC	REC	PRE	F1
PCA+SVM	0.53	0.52	0.64	0.56	0.60
DC+SVM	0.52	0.53	0.65	0.57	0.61
EC+SVM	0.55	0.53	0.69	0.57	0.62
CC+SVM	0.54	0.52	0.59	0.58	0.59
BC+SVM	0.53	0.50	0.66	0.56	0.61
CNN	0.58	0.62	0.62	0.65	0.60
GCN	0.57	0.63	0.50	0.64	0.55
ST-GCN	0.60	0.64	0.61	0.65	0.63
UCGL	0.63	0.65	0.60	0.68	0.65
Ours	0.66	0.66	0.68	0.70	0.69

boldface denotes the best results

TABLE II. Tested cross-site categorization: model from site 20 to site 1 on REST-MDD.

method	ACC	AUC	REC	PRE	F1
PCA+SVM	0.54	0.56	0.59	0.54	0.56
DC+SVM	0.57	0.61	0.59	0.57	0.58
EC+SVM	0.53	0.53	0.55	0.53	0.54
CC+SVMs	0.55	0.55	0.53	0.56	0.54
BC+SVM	0.52	0.52	0.59	0.52	0.55
CNN	0.58	0.59	0.55	0.61	0.56
GCN	0.58	0.66	0.54	0.61	0.54
ST-GCN	0.61	0.68	0.56	0.61	0.58
UCGL	0.63	0.66	0.60	0.62	0.63
Ours	0.64	0.65	0.71	0.70	0.67

A. Experiment Setting

We used five different metrics to evaluate the performance of the model, including accuracy (ACC), area of the curve (AUC) of subject features, recall (REC), precision (PRE), and F1 score (F1).

In this study, we consider cross-site classification, where models are pre-trained/trained on one imaging site from the fMRI dataset and fine-tuned on other sites from the same dataset. For MDD vs. HC classification on REST-MDD, we performed unsupervised pretraining of the model on site 20 and then fine-tuned and tested it on site 21 and site 1, respectively.

B. Methods for comparison

In this study we have conducted a comprehensive comparison of our proposed MCS-GL model with a suite of five conventional machine learning approaches, each pairing a distinct feature extraction technique (PCA, DC, EC, CC, and BC) with a Support Vector Machine (SVM) and three popular deep learning methods (CNN, GCN, and ST-GCN) to demonstrate the advantages of our self-supervised learning approach in situations where labeled data is scarce. PCA + SVM uses PCA for dimensionality reduction before SVM classification. DC + SVM measures node importance using Degree Centrality before SVM classification. EC + SVM leverages Eigenvector Centrality for influential node identification before SVM classification. CC + SVM uses Closeness Centrality to determine node proximity before SVM classification. BC + SVM quantifies node bridging frequency along shortest paths before SVM classification. For the REST-MDD experiment, the above methods were pre-trained on site 20 and tested or fine-tuned on sites 21 and 1.

C. Impact of Connection Thresholds

In our experiments, we used fMRI data and constructed brain network map data using a connectivity threshold by calculating the BOLD signal correlation between two brain regions; when the correlation is greater than or equal to the threshold, the two brain regions are connected. When the correlation is less than the threshold, the two brain regions are not connected. We set the connection threshold $\{0.1, 0.2, 0.3, 0.4, 0.5\}$ on site 20 to site 21 for comparison. From Fig. 1, it can be seen that as the connection threshold increases, the connections of the brain network increase, and the time complexity of the model increases, but the performance of the model does not correlate positively with the connection threshold, and the accuracy reaches its highest when the connection threshold reaches 0.3, which may be due to the fact that the complexity of the brain network is increased when the connection threshold is too small, which increases the level of noise in the raw data leading to the model having difficulty in extracting the effective features; and when the threshold is increased, the spatial features of the data are not well captured and some important connection information may be filtered out. It can be seen that the connection threshold is highly sensitive, and it is very important to choose an appropriate threshold for the construction of brain networks.

D. Impact of diffusion probability

In the experiment, we choose ST-GCN as the fMRI feature extractor. In order to study the effect of diffusion probability on the experimental results, we take the diffusion

probability $\{0.2, 0.3, 0.4, 0.5, 0.6\}$ for comparative tests to analyze the effect of different layers on the experimental results, and take site 20 to site 21 as an example. From Fig. 2, it can be seen that when the diffusion probability increases, the accuracy increases slowly; increasing the diffusion probability helps to improve the generalization ability of the model, which makes the model have a superior performance.

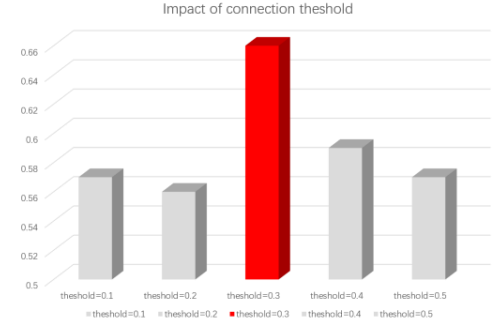


Fig. 1. Impact of connection threshold.

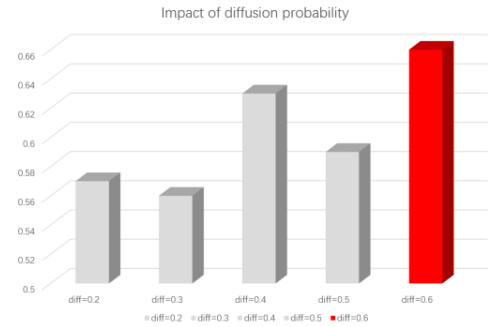


Fig. 2. Impact of diffusion probability

E. Influence of the projection layer

In our experiments, we used both the node projection layer and the graph projection layer in order to align the local and global features dimensionally. However, in our training, we trained both the node projection layer and the graph projection layer. However, the node projection layer is not used in the downstream classification task. Therefore, we consider the impact of training the node projection layer on the experiment. We compare the effect of training only the graph projection layer versus training both the node projection layer and the graph projection layer on the convergence of the model and the classification accuracy of the downstream task. As shown in Fig. 3, it can be seen from the figure species that the accuracy of the model decreases when both projection layers are trained at the same time, which may be due to the fact that when the projection layers are updated at the same time, the model has a higher degree of fit to the original data, such that the model is more ineffective in the target domain. In contrast, when updating only the graph projection layer, the model does not fit the data as well as when updating at the same time, and the prediction results for the target domain are

relatively better. Therefore, updating only the map projection layer during pre-training reduces overfitting and is more effective for fine-tuning on different data.

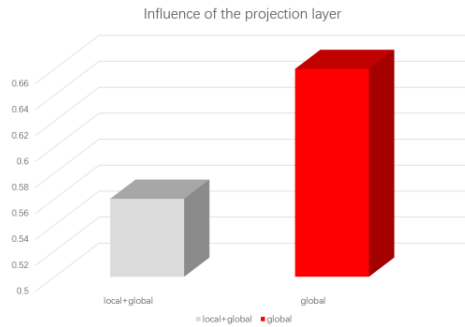


Fig. 3. Influence of the projection layer

IV. CONCLUSION

The small sample problem of clinical data becomes a bottleneck in the construction of brain network-based prior intelligent diagnosis and treatment models, in this paper, we propose a graph self-supervised learning framework, which consists of pre-training and fine-tuning, where the pre-training part generates semantically-rich representations using unlabeled functional magnetic resonance data to go on to perform the downstream tasks, and we set up a ppr-based graph diffusion method BOLD data to increase the sample size, and then we train the encoder by comparing the representations among the local global ones in unsupervised. The diagnosis of degenerative neurological diseases has always been a tricky problem in the clinic, and the self-supervised learning framework based on graph neural network proposed in this paper solves the small sample problem of fMRI data well and improves the accuracy of the diagnosis of neurological diseases. However, there are still many shortcomings. We summarize the whole paper to give some pending and future research works that need to be carried out, including:

1) Regarding the establishment of the data structure of brain network map, this paper only adopts to judge the connection between brain regions by calculating the correlation. The next step is to combine other properties between brain regions to build new data suitable for suitable deep learning frameworks: including but not limited to hypergraphs, etc., in order to capture deeper information between brain regions.

2) In this paper, we focus on encoding fMRI data using a temporal-spatial encoder and have chosen a graph self-supervised learning strategy. In our next work, we will explore whether there are better encoder choices for temporal-space data and better self-supervised learning strategies to improve classification performance.

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